

Please check the examination details below before entering your candidate information

Candidate surname						Other names					
Centre Number						Learner Registration Number					

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Wednesday 8 January 2025

Afternoon (Time: 40 minutes)

Paper reference **31617H/1B**

Applied Science/Forensic and Criminal Investigation

UNIT 1: Principles and Applications of Science I

Biology

SECTION A: STRUCTURES AND FUNCTIONS OF CELLS AND TISSUES

You must have:
a calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each:
 - Section A: Structures and functions of cells and tissues (Biology)
 - Section B: Periodicity and properties of elements (Chemistry)
 - Section C: Waves in communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☐. If you change your mind about an answer, put a line through the box ☐ and then mark your new answer with a cross ☐.

- 1 Figure 1 shows the membrane potential (mV) before, during and after an action potential in a neurone.

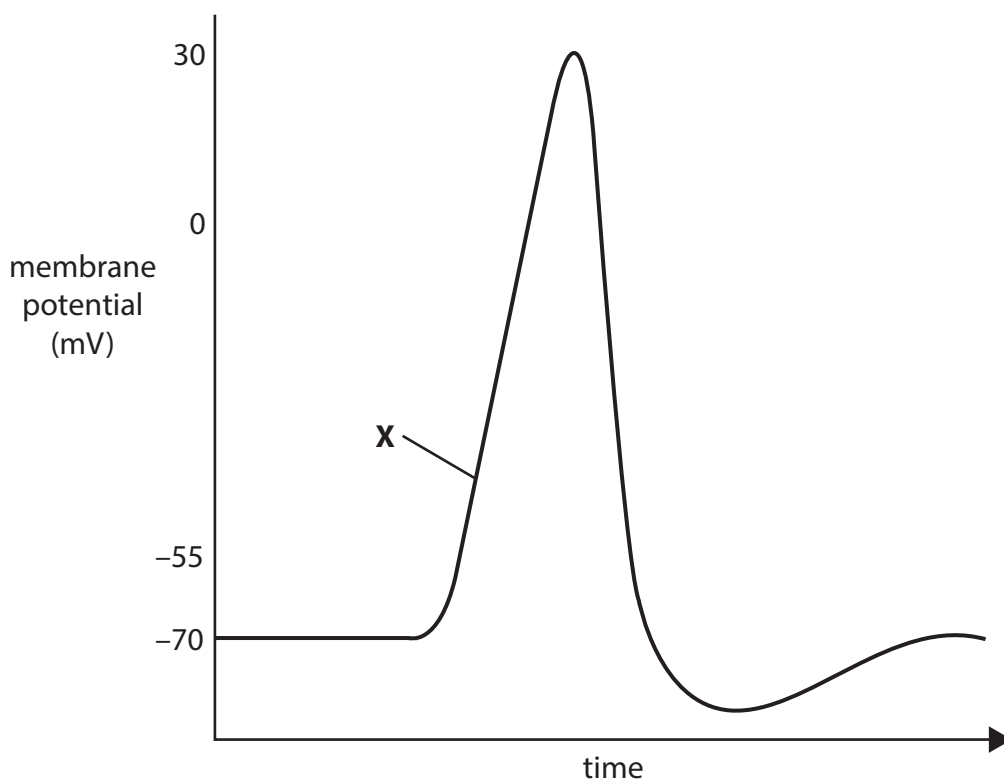


Figure 1

- (a) Identify the stage of the action potential labelled X in Figure 1.

(1)

- ☐ A depolarisation
☐ B hyperpolarisation
☐ C refractory period
☐ D repolarisation

- (b) State which ion enters the neurone during stage X of Figure 1.

(1)



(c) Identify the threshold membrane potential that must be reached to generate an action potential.

(1)

- ☐ **A** -70 mV
- ☐ **B** -55 mV
- ☐ **C** 0 mV
- ☐ **D** 30 mV

An action potential travels through an axon.

Axons can be myelinated or unmyelinated.

(d) Explain the effect of myelination on the speed of an action potential.

(2)

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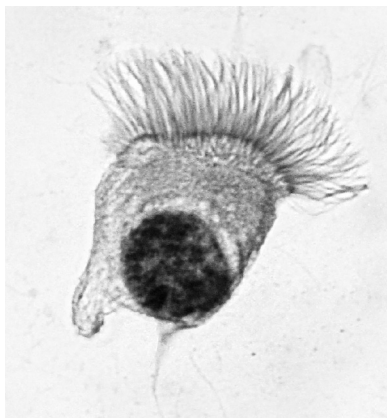
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(Total for Question 1 = 5 marks)



2 Figure 2 shows a ciliated epithelial cell.



(Source: © PAL)

Figure 2

A ciliated epithelial cell was observed through a light microscope.

The cell had an observed width of 3.2 mm.

The cell had an actual width of 12.8 μm .

(a) Calculate the magnification used to observe the cell.

(3)

Show your working.

magnification = $\times$



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(b) Explain why ciliated epithelial cells contain high numbers of mitochondria.

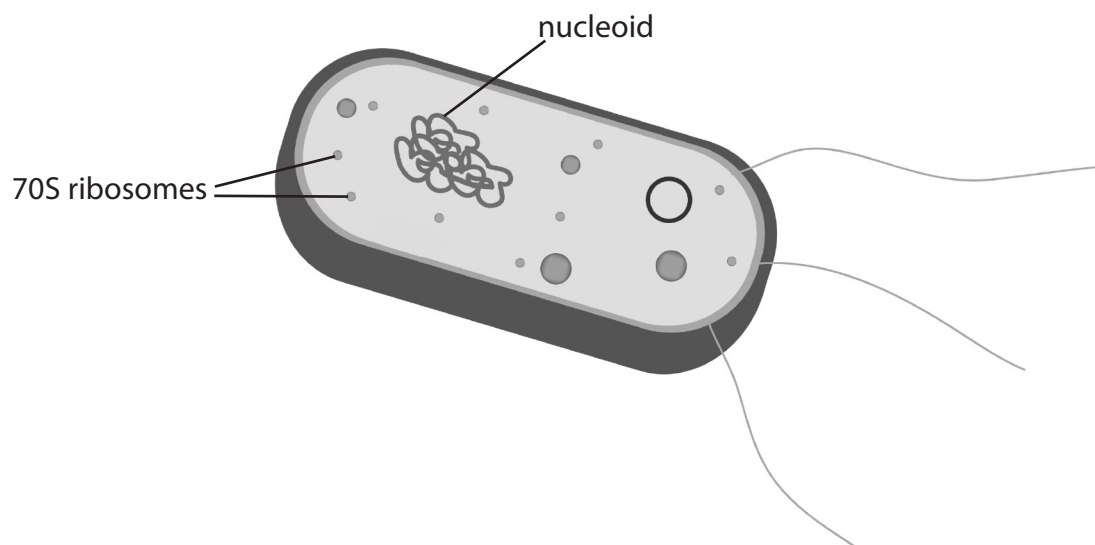
(3)

(Total for Question 2 = 6 marks)



P 8 0 4 7 5 A 0 5 1 2

3 Figure 3 shows a diagram of a prokaryote.



(Source: © PAL)

Figure 3

(a) State the function of the 70S ribosomes.

(1)

Plasmids are subcellular structures.

(b) Describe the structure of a plasmid.

(2)



An infection, caused by both Gram-negative and Gram-positive bacteria, is treated with antibiotics.

Figure 4 shows the number of bacterial cells in the infection.

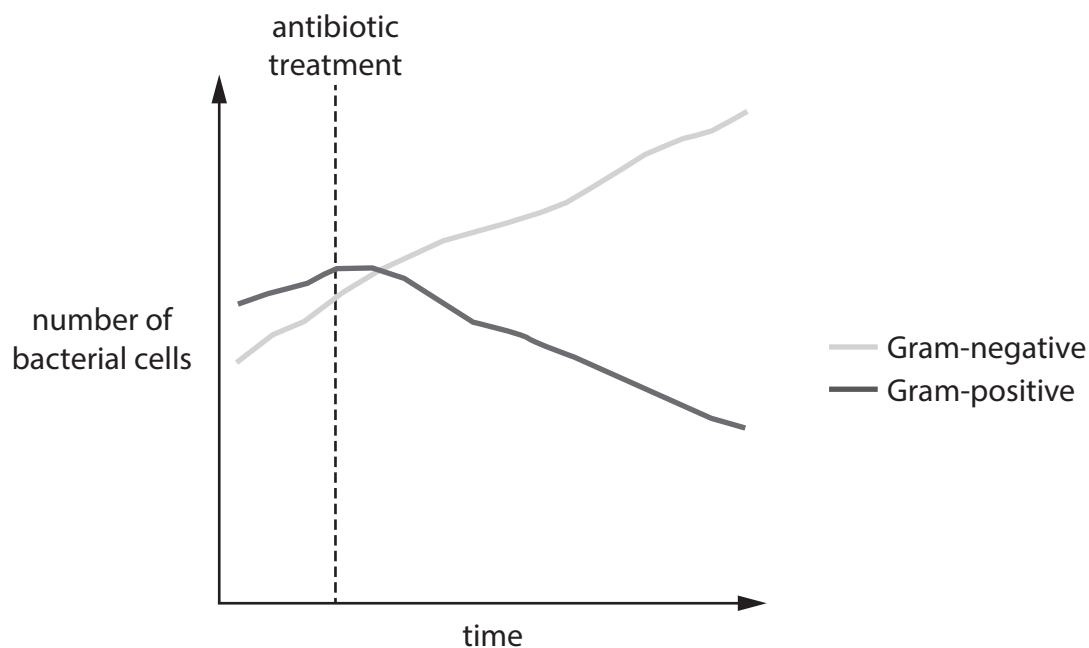


Figure 4

(c) Explain the effect of the antibiotic treatment.

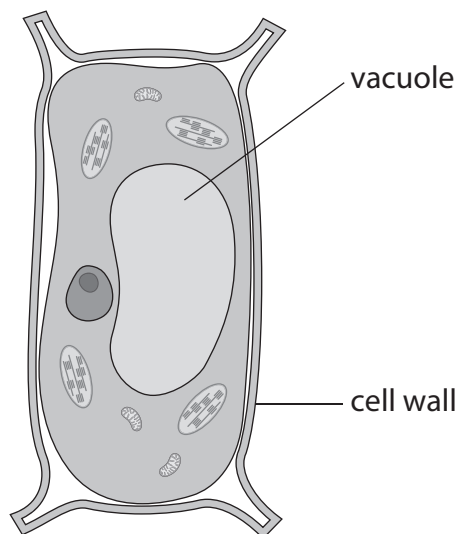
Use Figure 4 to support your answer.

(4)

(Total for Question 3 = 7 marks)



4 Figure 5 shows a palisade mesophyll cell from a plant.



(Source: © PAL)

Figure 5

- (a) Describe how the cell wall prevents the cell from bursting as the vacuole absorbs water.

(2)

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- (b) Explain how a large vacuole maximises the rate of photosynthesis.

(4)

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(Total for Question 4 = 6 marks)



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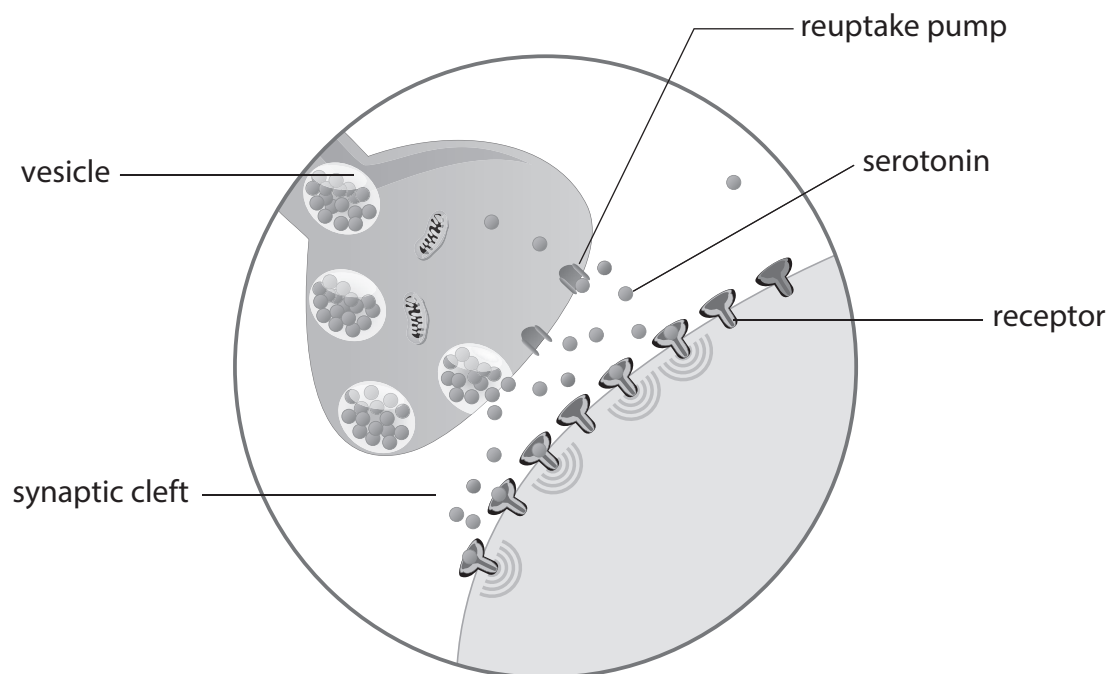
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QUESTION 5 STARTS ON NEXT PAGE



5 Serotonin is a neurotransmitter found in synapses in the brain.

Low levels of serotonin prevent transmission of impulses at synapses, leading to depression.

Figure 6 shows a diagram of a synapse.



(Source: © PAL)

Figure 6

Antidepressant drugs can be prescribed by a doctor to reduce the symptoms of depression in a patient.

Discuss different ways prescribed drugs could act at a synapse to treat depression.

Use the information in Figure 6 to support your answer.

(6)

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TOTAL FOR PAPER = 30 MARKS
TOTAL FOR EXAMINATION = 90 MARKS



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